

1.5 Solving Quadratic Equations – Part 1: Solve by Factoring – Worksheet

MCR3U

1) Solve by factoring

a) $x^2 + 8x + 12 = 0$

$$(x+6)(x+2) = 0$$

$$x+6=0$$

$$x = -6$$

$$x+2=0$$

$$x = -2$$

b) $h^2 + 9h + 18 = 0$

$$(h+6)(h+3) = 0$$

$$h+6=0$$

$$h = -6$$

$$h+3=0$$

$$h = -3$$

c) $m^2 + 3m = 0$

$$m(m+3) = 0$$

$$m = 0$$

$$m+3=0$$

$$m = -3$$

d) $w^2 - 18w + 56 = 0$

$$(w-14)(w-4) = 0$$

$$w-14=0$$

$$w = 14$$

$$w-4=0$$

$$w = 4$$

e) $x^2 - 2x = 0$

$$x(x-2) = 0$$

$$x = 0$$

$$x-2=0$$

$$x = 2$$

f) $c^2 - 17c + 30 = 0$

$$(c-15)(c-2) = 0$$

$$c-15=0$$

$$c = 15$$

$$c-2=0$$

$$c = 2$$

2) Solve

a) $3x^2 + 28x + 9 = 0$

S: 28
P: 27

27 and 1

$$3x^2 + 27x + 1x + 9 = 0$$

$$(3x^2 + 27x) + (1x + 9) = 0$$

$$3x(x+9) + 1(x+9) = 0$$

$$(x+9)(3x+1) = 0$$

$$x+9=0$$

$$x = -9$$

$$3x+1=0$$

$$x = -1/3$$

b) $4k^2 + 19k + 15 = 0$

S: 19
P: 60

4 and 15

$$4k^2 + 4k + 15k + 15 = 0$$

$$(4k^2 + 4k) + (15k + 15) = 0$$

$$4k(k+1) + 15(k+1) = 0$$

$$(k+1)(4k+15) = 0$$

$$k+1=0$$

$$k = -1$$

$$4k+15=0$$

$$k = -15/4$$

d) $16b^2 - 1 = 0$ D.O.S.

$$(4b-1)(4b+1) = 0$$

$$4b-1=0$$

$$b = 1/4$$

$$4b+1=0$$

$$b = -1/4$$

f) $4x^2 - 12x + 9 = 0$ P.S.T. $a^2 - 2ab + b^2 = (a-b)^2$

$$(2x-3)^2 = 0$$

$$2x-3=0$$

$$x = 3/2$$

3) Solve each quadratic equation by factoring

a) $x^2 + 2x - 3 = 0$ S: 2 P: -3 (3 and -1)

$$(x+3)(x-1) = 0$$

$$x+3=0 \quad x-1=0$$

$$x = -3 \quad x = 1$$

b) $x^2 + 3x - 10 = 0$ S: 3 P: -10 (5 and -2)

$$(x+5)(x-2) = 0$$

$$x+5=0 \quad x-2=0$$

$$x = -5 \quad x = 2$$

c) $4x^2 - 36 = 0$ D.O.S.

$$(2x-6)(2x+6) = 0$$

$$2x-6=0$$

$$x = 3$$

$$2x+6=0$$

$$x = -3$$

d) $6x^2 - 14x + 8 = 0$

$$2(3x^2 - 7x + 4) = 0$$

$$3x^2 - 7x + 4 = 0$$

$$3x^2 - 3x - 4x + 4 = 0$$

$$(3x^2 - 3x) + (-4x + 4) = 0$$

$$3x(x-1) - 4(x-1) = 0$$

$$(3x-4)(x-1) = 0$$

$$x_1 = \frac{4}{3} \quad x_2 = 1$$

f) $6x^2 + 19x + 10 = 0$

$$6x^2 + 15x + 4x + 10 = 0$$

$$(6x^2 + 15x) + (4x + 10) = 0$$

$$3x(2x+5) + 2(2x+5) = 0$$

$$(2x+5)(3x+2) = 0$$

$$2x+5=0$$

$$x = -5/2$$

$$3x+2=0$$

$$x = -2/3$$

e) $15x^2 - 8x + 1 = 0$ S: -8 P: 15 (-5 and -3)

$$15x^2 - 5x - 3x + 1 = 0$$

$$(15x^2 - 5x) + (-3x + 1) = 0$$

$$5x(3x-1) - 1(3x-1) = 0$$

$$(3x-1)(5x-1) = 0$$

$$3x-1=0 \quad 5x-1=0$$

$$x = 1/3$$

$$x = 1/5$$

P: 12 S: -7 (-3 and -4)

S: 19 P: 60 (15 and 4)

4) Solve by factoring

a) $-x^2 - 10x - 16 = 0$

$$-1(x^2 + 10x + 16) = 0$$

$$x^2 + 10x + 16 = 0$$

$$(x+2)(x+8) = 0$$

$$x+2=0 \quad x+8=0$$

$$x=-2 \quad x=-8$$

b) $6d^2 + 15d = -9$

$$6d^2 + 15d + 9 = 0$$

$$3(2d^2 + 5d + 3) = 0$$

$$2d^2 + 5d + 3 = 0$$

$$2d^2 + 2d + 3d + 3 = 0$$

$$(2d^2 + 2d) + (3d + 3) = 0$$

$$2d(d+1) + 3(d+1) = 0$$

$$(d+1)(2d+3) = 0$$

$$\begin{aligned} d+1 &= 0 & 2d+3 &= 0 \\ d &= -1 & d &= -\frac{3}{2} \end{aligned}$$

5) A rectangle has dimensions $x+10$ and $2x-3$. Determine the value of x that gives an area of 54 cm^2

$$(x+10)(2x-3) = 54$$

$$2x^2 - 3x + 20x - 30 = 54$$

$$2x^2 + 17x - 84 = 0$$

$$2x^2 + 24x - 7x - 84 = 0$$

$$(2x^2 + 24x) + (-7x - 84) = 0$$

$$2x(x+12) - 7(x+12) = 0$$

$$(x+12)(2x-7) = 0$$

~~$$\begin{aligned} x+12 &= 0 \\ x &= -12 \end{aligned}$$~~

$$2x-7=0$$

$$x = \frac{7}{2} \text{ or } 3.5 \text{ cm}$$

Answers

1) a) -2, -6 b) -3, -6 c) 0, -3 d) 14, 4 e) 0, 2 f) 15, 2

2) a) $\frac{-1}{3}, 9$ b) $-1, \frac{-15}{4}$ d) $\frac{1}{4}, \frac{-1}{4}$ f) $\frac{3}{2}$

3) a) -3, 1 b) -5, 2 c) -3, 3 d) $\frac{4}{3}, 1$ e) $\frac{1}{5}, \frac{1}{3}$ f) $-\frac{5}{2}, -\frac{2}{3}$

4) a) -8, -2 b) $-1, \frac{-3}{2}$

5) 3.5

